



EMPLOYEE LEAVE AND PAYROLL MANAGAMENT SYSTEM – Deliverable 1

Course: **Software Engineering**

Instructor: **Ms. Rabail Zahid**

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BY:

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INTRODUCTION

The Employee Leave and Payroll Management System is a Java-based business management application designed to automate employee record handling, leave management, and payroll processing. The system improves organizational efficiency by reducing manual work and ensuring accurate record keeping.

PROJECT VISION

To develop a reliable business system that simplifies HR operations by providing automated employee management, leave processing, and payroll calculation.

INTENDED USERS

Stakeholders

- HR Manager
- Employees
- Organization Management

Stakeholder Needs

- Maintain employee records
- Manage leave requests
- Generate payroll
- Track attendance

FEATURES / FUNCTIONALITY

- Employee registration
- Leave request management
- Leave approval system
- Salary management
- Payroll calculation
- Record viewing

System helps by:

- Reducing paperwork
- Automating calculations
- Improving accuracy

USER STORIES

US-01 - Login to System

As an HR Admin **I want** to login using valid credentials **So that** I can securely access the system

US-02 - Add Employee Record

As an HR Admin I want to add a new employee So that employee data is stored in the system

US-03 - View Employee Records

As an HR Admin I want to view employee list So that I can manage employee data

US-04 - Logout from System

As a System User I want to logout securely So that unauthorized access is prevented

Employee Management

US-05 - Update Employee Record

As an HR Admin, I want to update employee information so that records remain accurate.

US-06 - Delete Employee Record

As an HR Admin, I want to remove employee records so that inactive employees are not stored.

Attendance Management

US-07 - Record Employee Attendance

As an HR Admin, I want to mark attendance so that employee presence is tracked.

US-08 - View Attendance History

As an HR Admin, I want to view attendance reports so that I can monitor employee performance.

Leave Management

US-09 - Apply for Leave

As an Employee, I want to submit leave requests so that I can request time off.

US-10 - Approve/Reject Leave

As an HR Admin, I want to approve or reject leave requests so that leave records remain controlled.

US-11 - View Leave History

As an Employee, I want to view my leave status so that I know approval results.

Payroll Management

US-12 - Generate Payroll

As an HR Admin, I want to calculate employee salaries so that payroll is automated.

US-13 - View Payslip

As an Employee, I want to view my payslip so that I know my salary details.

Advance Payment Management

US-14 - Record Salary Advance

As an HR Admin, I want to record advance payments so that deductions can be applied later.

US-15 - Deduct Advance from Salary

As an HR Admin, I want advance payments deducted automatically so that payroll is accurate.

Banking & Transactions

US-16 - Process Bank Transaction

As a Finance Officer, I want to transfer salaries to bank accounts so that employees receive payment.

Reporting & Compliance

US-17 - Generate Tax Report

As a Finance Officer, I want tax reports generated so that compliance requirements are met.

US-18 - View Audit Logs

As an Admin, I want to view system logs so that system activities are traceable.

Access Control

US-19 - Manage User Roles

As an Admin, I want to assign roles so that access is controlled.

US-20 - Manage Permissions

As an Admin, I want to define access rights so that users can only perform authorized actions.

STRUCTURED SPECIFICATIONS

1. Login User

Inputs: username, password

Process:

1. Validate input
2. Check database
3. Create session

Output: dashboard / error

2. Add Employee

Inputs: name, ID, salary, department

Process:

- i. Validate fields
- ii. Store in database

Output: success message

3. View Employees

Inputs: request from admin

Process: fetch all records

Output: employee list

4. Logout

Inputs: logout request

Process: destroy session

Output: login page

5. Update Employee Record

Inputs: employee ID, updated details

Process:

- i. Validate input data
- ii. Locate employee in database
- iii. Update record

Output: update confirmation message

6. Delete Employee Record

Inputs: employee ID

Process:

- i. Verify employee exists
- ii. Remove record from database

Output: deletion confirmation message

7. Record Attendance

Inputs: employee ID, date, status

Process:

- i. Validate employee ID
- ii. Store attendance record

Output: attendance recorded message

8. View Attendance History

Inputs: employee ID / date range

Process:

- i. Retrieve attendance records
- ii. Display filtered results

Output: attendance report

9. Apply for Leave

Inputs: employee ID, leave type, dates, reason

Process:

- i. Validate request details
- ii. Store leave request
- iii. Mark status as pending

Output: leave submission confirmation

10. Approve/Reject Leave

Inputs: leave request ID, decision

Process:

- i. Retrieve leave request
- ii. Update approval status

Output: decision confirmation

11. View Leave History

Inputs: employee ID

Process:

- i. Retrieve leave records
- ii. Display status history

Output: leave history list

12. Generate Payroll

Inputs: employee records, attendance, salary data

Process:

- i. Calculate salary
- ii. Apply deductions
- iii. Store payroll record

Output: payroll generated message

13. View Payslip

Inputs: employee ID, pay period

Process:

- i. Retrieve payroll data
- ii. Display payslip details

Output: payslip

14. Record Advance Payment

Inputs: employee ID, amount, date

Process:

- i. Validate payment details
- ii. Store advance record

Output: advance recorded message

15. Deduct Advance from Salary

Inputs: employee ID, payroll period

Process:

- i. Retrieve advance records
- ii. Deduct from salary
- iii. Update payroll

Output: deduction confirmation

16. Process Bank Transaction

Inputs: employee bank details, salary amount

Process:

- i. Validate account details

- ii. Transfer salary
- iii. Record transaction

Output: transaction confirmation

17. Generate Tax Report

Inputs: payroll data, tax rules

Process:

- i. Calculate tax deductions
- ii. Generate report

Output: tax report

18. View Audit Logs

Inputs: system request

Process:

- i. Retrieve activity logs
- ii. Display chronological records

Output: audit log list

19. Manage User Roles

Inputs: user ID, role type

Process:

- i. Validate role
- ii. Assign role to user

Output: role assignment confirmation

20. Manage Permissions

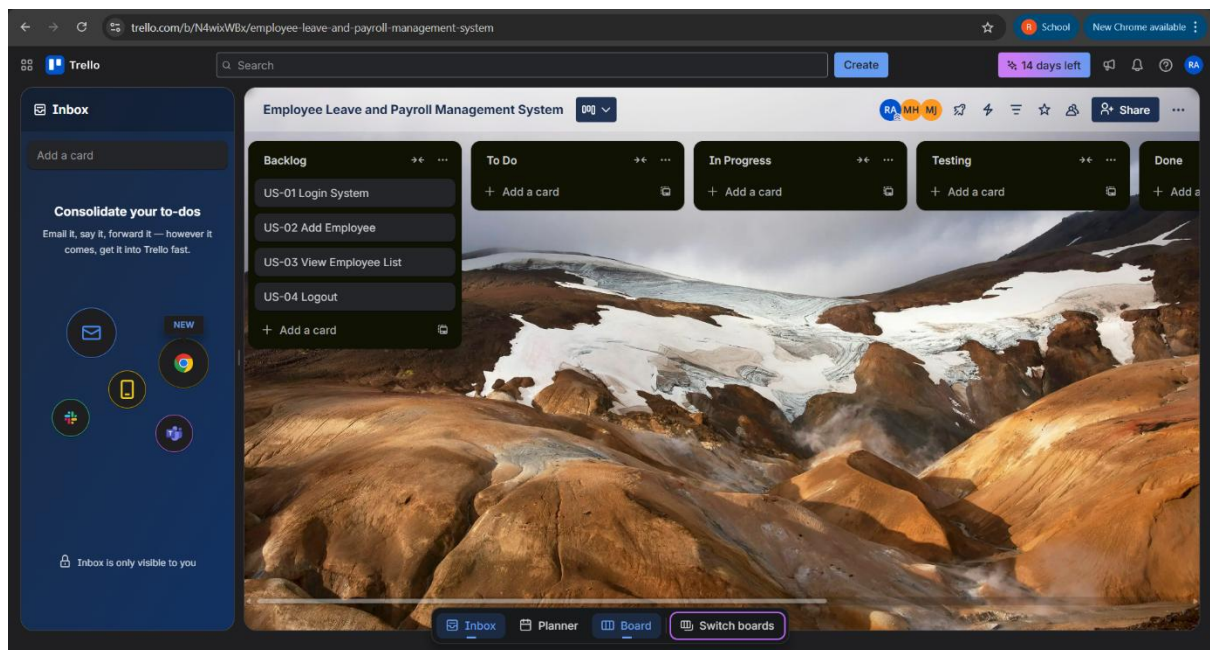
Inputs: role ID, permission settings

Process:

- i. Update access rights
- ii. Save configuration

Output: permission update confirmation

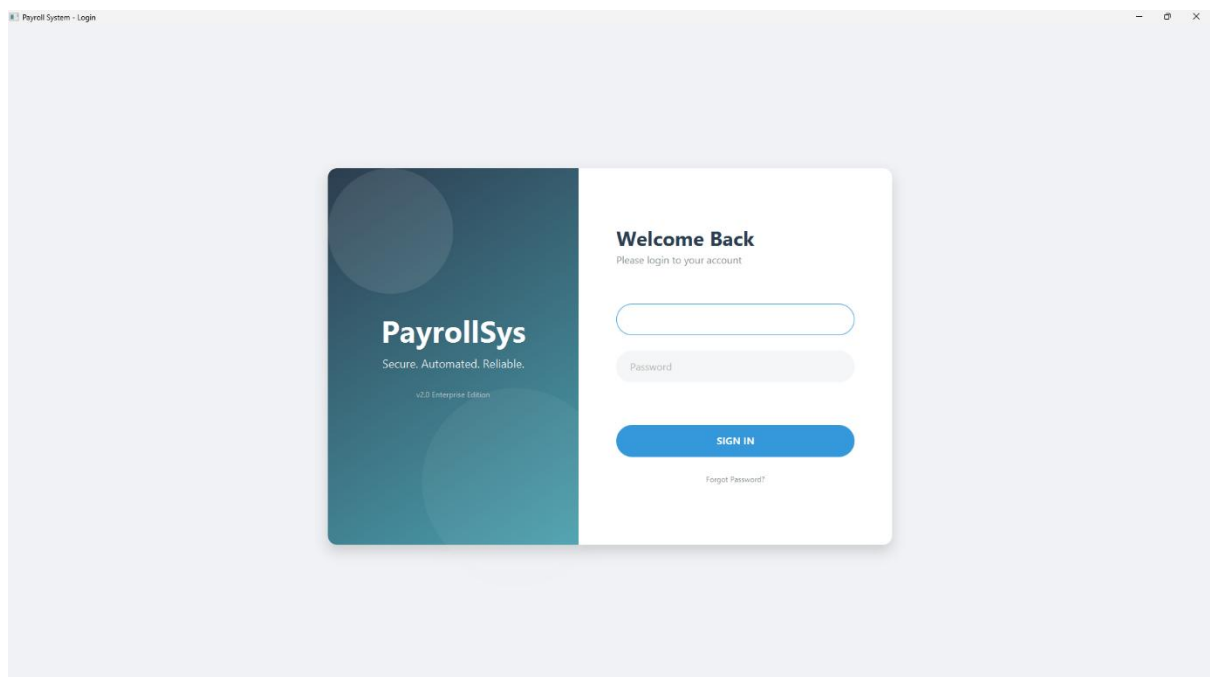
SCRUM BOARD

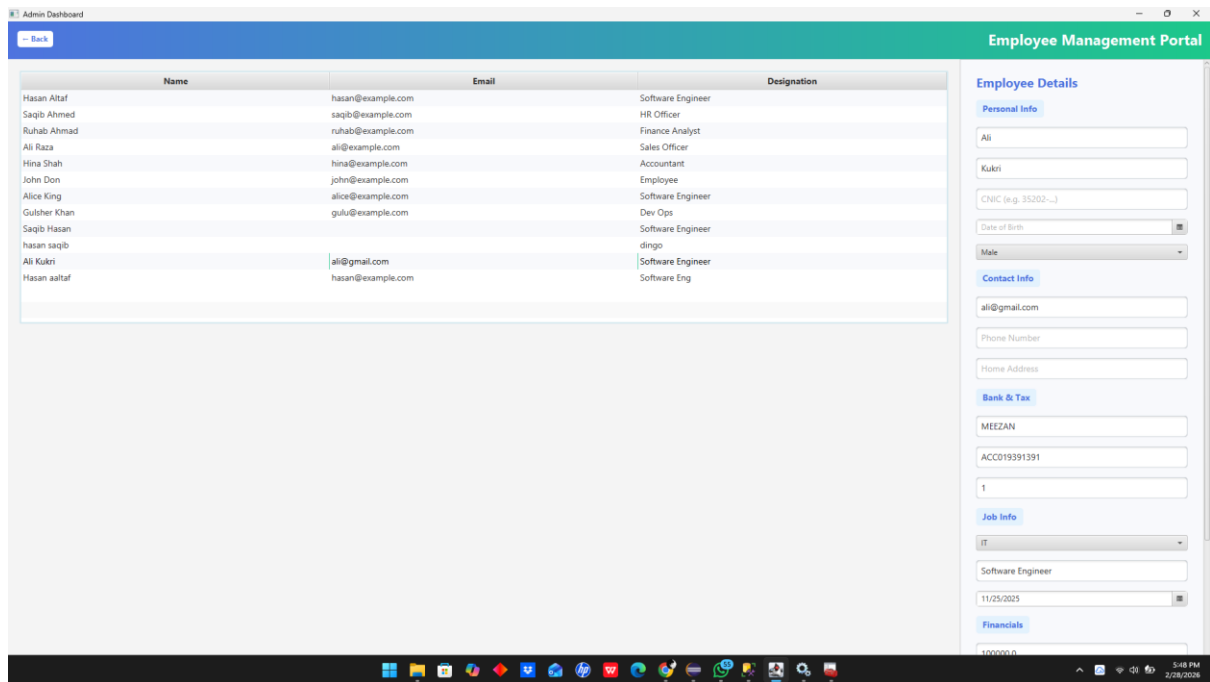


NFR (NON-FUNCTIONAL REQUIREMENTS)

- System must respond within 2 seconds
- Data must be stored securely
- System must be user-friendly
- System must support multiple users
- System must be reliable

ITERATION 1 IMPLEMENTATION SCREENSHOTS





WORK DIVISION

Task	Responsible
Requirements	Ruhab
UI Design	Saqib
Implementation	Ruhab + Saqib
Scrum Management	Hasan
Testing	All members